

Cosmology & Astronomical Instrumentation

Problem Set

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This problem set accompanies the *Cosmology & Astronomical Instrumentation* handout. Section A has five objective (multiple-choice) questions; Section B has three subjective questions, pitched a little harder than Section A. Use $c = 3 \times 10^8$ m/s, $G = 6.674 \times 10^{-11}$ N m²kg⁻², $m_H = 1.673 \times 10^{-27}$ kg, 1 rad = 206265'', and 1 Mpc = 3.086×10^{22} m where needed, unless a question tells you otherwise.

Section A — Objective

O1. A telescope with $D = 10$ m has, compared to one with $D = 2$ m, a light-gathering power greater by a factor of:

- (a) 5
- (b) 10
- (c) 25
- (d) 50

O2. The brightness of a star's image (a point source, as opposed to an extended object) in a telescope depends on:

- (a) the focal ratio $f/\#$ only
- (b) the aperture diameter D only
- (c) both D and $f/\#$ equally
- (d) neither — stars appear equally bright through any telescope

O3. A telescope has aperture $D = 1.0$ m, observing at $\lambda = 550$ nm. Its diffraction-limited resolution θ is closest to:

- (a) 0.14''
- (b) 1.4''
- (c) 0.014''
- (d) 14''

O4. Which one of the following statements about the cosmic distance ladder is **false**?

- (a) Parallax distances require no external calibration.
- (b) Cepheid distances are calibrated using nearby Cepheids with known parallax distances.
- (c) Type Ia supernovae are calibrated completely independently of Cepheids, using only their own light curves.
- (d) An unnoticed systematic error in the Cepheid rung would bias every rung built on top of it.

O5. A photon was emitted when the universe's scale factor was $a_{\text{then}} = 0.25$. Its observed redshift z is:

- (a) 3
- (b) 4
- (c) 0.25
- (d) 0.75

Section B — Subjective

S1. An observer images a faint galaxy with a CCD. The source count rate (summed over its photometric aperture) is $N_{\star} = 5.0$ electrons/s, spread over $n_{\text{pix}} = 25$ pixels, with sky background $N_{\text{sky}} = 20$ electrons/s/pixel, dark current $N_{\text{dark}} = 2$ electrons/s/pixel, and read noise $R = 8$ electrons/pixel.

- (i) Find the SNR for an exposure time $t = 100$ s.
- (ii) Find the SNR for $t = 1000$ s. Compare the growth in SNR between the two exposures to what you'd expect in the purely signal-dominated regime ($\text{SNR} \propto t$) versus the background-dominated regime ($\text{SNR} \propto \sqrt{t}$), and state which regime this observation is actually in.
- (iii) Estimate the exposure time t needed to reach $\text{SNR} = 10$.

S2. Two radio dishes, each of diameter $D = 12$ m, are combined into a VLBI interferometer with baseline $B = 8000$ km (comparable to the real Event Horizon Telescope), observing at $\lambda = 1.3$ mm.

- (i) Find the interferometric angular resolution $\theta_{\text{min}} \sim \lambda/B_{\text{max}}$, in microarcseconds (μ'').
- (ii) Find the diffraction-limited resolution of a single one of the 12 m dishes alone, at the same wavelength, and find the factor by which interferometry improves on it.
- (iii) The M87* black hole shadow imaged by the EHT has an angular diameter of about $42 \mu''$. Using your answer from (i), comment on whether this baseline is capable of resolving the shadow at all, and by roughly what factor.

S3. A galaxy is observed with redshift $z = 0.05$.

- (i) Using Hubble's Law with $H_0 = 70 \text{ km/s/Mpc}$ (valid since $z \ll 1$), estimate the galaxy's distance in Mpc.
- (ii) Find the scale factor a_{then} at the time this galaxy's light was emitted, relative to today ($a_{\text{now}} = 1$).
- (iii) Convert $H_0 = 70 \text{ km/s/Mpc}$ to SI units, and use it to estimate the critical density ρ_c of the universe today, in kg/m^3 . Then express this as an equivalent number of hydrogen atoms per cubic metre. Briefly comment on why such a tiny density threshold is nonetheless cosmologically significant.